

Executive Summary

Virtual Cell (VCell) is a freely-available, open-source framework for modeling and simulating cellular processes. Developed at UConn's Center for Cell Analysis and Modeling (CCAM) since the late 1990s, VCell provides a user-friendly web interface and client application to build biochemical/electrophysiological models and run simulations (deterministic, stochastic, spatial) without requiring users to write code ¹ ² . The project has been continuously funded by NIH (initially NCRR, now NIGMS R24 GM137787 ² ³) and has produced a suite of tools (BioModel, MathModel, Geometry, rule-based modeling, ImageJ integration, etc.) to translate biological hypotheses into mathematical systems automatically ¹ ⁴ . Over 25 years, key milestones include the first VCell modeling papers (1997–2002), major software releases (e.g. VCell 4.0 in 2005, 5.0 in 2011, 7.0 in 2018), and many enhancements (hybrid stochastic solvers, rule-based modeling, SBML/COMBINE support). VCell's impact is documented in dozens of publications (notably **Schaff et al. 1997 Biophys J**; **Moraru et al. 2008 IET Syst Biol**; **Resasco et al. 2012 WIREs**; **Blinov et al. 2017 Biophys J**; **Schaff et al. 2016 PLoS Comp Biol**). The CCAM/VCell team (Loew, Schaff, Slepchenko, Moraru, Blinov, et al.) leads development and training, and the open-source repository on GitHub (MIT license) fosters community contributions (e.g. PyVCell, VCell-Fiji plugin, VCell-AI). VCell integrates standards and tools (SBML, SED-ML, BioPAX/Pathway Commons, COPASI, Smoldyn, BioNetGen) and has been applied to modeling calcium dynamics, chemotaxis, FRAP experiments, and many cell-specific case studies. Below is a detailed chronological account (with a timeline table) of VCell's development, major features, publications, and ecosystem, followed by a suggested slide presentation outline.

What is Virtual Cell?

VCell is a **computational modeling environment** for cell biology. It allows users to define *BioModels* (biological descriptions of species, reactions, compartments) and *MathModels* (the resulting equations) and then run simulations under various assumptions (deterministic ODEs/PDEs, stochastic algorithms, hybrid approaches) ¹ ⁵ . The system is distributed (client-server architecture) so users can build complex models via a graphical Java interface while heavy simulations run on servers. Its unique “three-layer abstraction” (Physiology → Mathematical description → Numerical simulation) separates the biological hypothesis from the computational details ¹ ⁶ . VCell's features include:

- **Compartments & Geometry:** Define nested/multicompartment models and incorporate real 2D/3D cell geometries (e.g. from microscope images) into spatial simulations ⁵ ⁷ .
- **Solvers:** ODE solvers for well-mixed models, stochastic (Gillespie) solvers, particle-based (e.g. Smoldyn) for discrete spatial modeling, and hybrid solvers that mix deterministic and stochastic dynamics ⁵ ⁸ .
- **Rule-Based Modeling:** Integrated support for rule-based modeling of molecular interactions (via BioNetGen/NFSim) so that complex binding/unbinding can be encoded without enumerating all species ⁹ ¹⁰ .
- **Data Integration:** Upload experimental data/images as *Field Data* for simulations (e.g. FRAP modeling), link species to databases (UniProt, PubMed), import SBML/CellML models, and query pathway databases (BioPAX/Pathway Commons) within VCell ¹¹ .
- **Collaboration & Sharing:** Centralized database at UConn stores user models; models can be kept private, shared, or made public. Over 300 published models are available online (with DOI/PubMed links) ¹² .
- **Modularity & Extensibility:** VCell is open source under the MIT license, with components (vcell-core,

client, server, solvers, etc.) on GitHub ¹³ ¹⁴ . Community tools include a Python API (PyVCell) and an ImageJ (Fiji) plugin for integrating image analysis with VCell simulations ¹⁵ ¹⁶ .

In summary, VCell provides “one-stop” modeling of cell-biological systems, from construction to computation, aimed at bridging experimental and theoretical biology ¹ ¹⁷ . Users range from bench biologists to computational modelers; no deep programming is required for standard use.

Timeline of VCell Milestones

Year	Event	Source
1994	Center for Cell Analysis and Modeling (CCAM) established at UConn (diverse team develops VCell) ¹⁸ .	CCAM/UConn Website ¹⁸
1997	First VCell paper: “ <i>A general computational framework for modeling cellular structure and function</i> ” (Schaff et al., Biophys J) ¹⁹ .	Schaff et al. 1997 ¹⁹
1998	IFIP Conf. paper on VCell framework (Shin et al., 1998) – introduced VCell concept (Proc. IFIP VDBS) ²⁰ .	Shin et al. 1998 ²¹
1999	PSB Conference: “ <i>The Virtual Cell</i> ” (Schaff & Loew) – early VCell system description ²² . NIH supports VCell via Biomedical Technology Resource (NCRR grant RR13186).	Loew & Schaff 1999 (PSB); NIH Grant ²² ³
2001	<i>Trends Biotech</i> review of VCell (Loew & Schaff, 2001) describing features and applications.	Loew & Schaff 2001 ²³
2002	Ann. NY Acad. Sci. paper “The virtual cell: an integrated modeling environment...” (Moraru et al.) ²⁴ . VCell project recognized at Winter Simulation Conf.	Moraru et al. 2002 ²⁴
2005	VCell 4.0 released (completely redesigned UI; multiple model editing; database manager) ²⁵ .	VCell Release Notes ²⁵
2008	VCell 4.4/4.5 released (annotations, SBML enhancements, hybrid stochastic sims) ²⁶ ²⁵ . IET <i>Systems Biology</i> paper “Virtual Cell modeling environment” (Moraru et al.) ¹ ²⁷ .	VCell 4.4/4.5 Release Notes; Moraru et al. 2008 ¹
2011	VCell 5.0 released (major GUI overhaul; image segmentation; Smoldyn stochastic solver; BioPAX interface) ⁷ .	VCell Release Notes ⁷
2012	WIREs <i>Syst Biol</i> review “Virtual Cell: computational tools for modeling...” (Resasco et al. 2012) ²⁸ . Methods Cell Biol. chapter on spatial modeling with VCell (Cowan et al. 2012) ²⁹ .	Resasco et al. 2012 ²⁸
2014	VCell integrates Pathway Commons/BioPAX queries (Blinov et al. 2014 Bioinformatics) ³⁰ .	Blinov et al. 2014 ³⁰
2016	VCell 6.0 (rule-based modeling support) features introduced. PLoS Comput Biol <i>hybrid solver</i> paper (Schaff et al. 2016) ⁸ .	Schaff et al. 2016 ⁸

Year	Event	Source
2017	VCell 6.1 released (GUI for rule-based models) ³¹ . Biophys J paper on VCell rule-based modeling (Blinov et al. 2017) ³² .	Blinov et al. 2017 ³²
2018	VCell 7.0 released (new architecture; 2D kinematics; UI enhancements) ³³ .	VCell Release Notes ³³
2019–2021	VCell 7.x series (7.1–7.4) added COMBINE/SBML-SEDML support, containerized solvers, GUI and performance improvements ³⁴ ³⁵ .	VCell Release Notes ³⁴
2023	VCell 7.5 released (expanded SBML/SEDML import/export, COPASI integration, CLI tools) ³⁶ .	VCell Release Notes ³⁶
2025	VCell 7.7 released (Feb 2026) – installer no longer needs admin rights ³⁷ . Recent case studies published (e.g. FRAP in Bacillus spores, EGFR-Raf signaling, Tau/microtubule dynamics ³⁸ ³⁹).	VCell News ³⁷ ³⁹

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timeline
  title Virtual Cell (VCell) Key Milestones
  1994 : CCAM founded at UConn 18
  1997 : First VCell paper (Biophys J) 19
  2002 : VCell ANNYAS env. paper 24
  2005 : VCell 4.0 release 25
  2011 : VCell 5.0 release 7
  2018 : VCell 7.0 release 33
  2021 : VCell 7.4 release 34
  2023 : VCell 7.5 release 36
  2026 : VCell 7.7 release (Feb 2026) 37

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Core Developers, Institutions, and Funding

The VCell project is housed at **UConn Health's Center for Cell Analysis and Modeling (CCAM)**, led by Prof. Leslie Loew. CCAM was founded in 1994 and brings together biologists, physicists, chemists and software engineers to develop computational tools ¹⁸ . Key developers include **James Schaff, Boris Slepchenko, Ion Moraru, Michael Blinov** and others ⁴⁰ ¹ . VCell has been funded primarily by NIH: originally under NCRR (grant RR13186) as a Biomedical Technology Resource, and currently by NIGMS (grant R24 GM137787) ² ³ . CCAM's resources (microscopy and HPC clusters) support VCell development and user training. The GitHub organization *virtualcell* (Virtual Cell) hosts the code (client, server, solvers) under an open-source MIT license ¹³ ¹⁶ . In 2019 UConn received a National Resource grant (R24 GM137787) explicitly to maintain VCell and COPASI as NIH-supported modeling resources ² .

Architecture and Major Features

VCell's architecture comprises a **client GUI**, a **model database server**, and **solver backends**. Models are built in the **BioModel** interface (biological view) or **MathModel** interface (equations view) ⁴¹ ⁴² . A *Physiology* specification defines species, reactions, and compartment topology; *Applications* within a model

specify simulation conditions and solver types ¹ ⁴³. VCell automatically generates the mathematical representation (VCML) and dispatches it to numerical solvers.

Key modules include:

- **Solver Suite:** CVODE/IDA for ODEs, custom finite-volume PDE solver, Gillespie stochastic solvers, NFSim (network-free) via BioNetGen integration, Smoldyn for particle-based reaction-diffusion ¹ ⁹. A hybrid spatial solver (mixing stochastic and deterministic) was added (Schaff et al. 2016) ⁸.
- **Spatial Geometry:** A **Geometry** editor lets users create/import 1D/2D/3D shapes (via templates, math expressions, or image segmentation) ¹¹ ²⁵. Realistic cell geometries from microscopy can be imported so diffusion and transport occur in true geometry ⁵.
- **Rule-Based Modeling:** VCell 6.x introduced full support for rule-based models. Users can graphically specify molecule structures and rules; VCell uses BioNetGen/NFSim to simulate, allowing efficient representation of large signaling networks ⁹ ⁴⁴.
- **Data and Standards:** VCell supports SBML and CellML import/export. As of VCell 7.x, it fully supports COMBINE archives (.omex) including SED-ML for experiments ³⁴. The DB uses MIRIAM annotations linking model entities to public databases (PubMed, UniProt, etc.) ¹⁵. Users can query Pathway Commons or SABIO-RK from VCell to fetch existing model components ¹¹. A *Field Data* feature uses images or concentration maps as spatial initial conditions (e.g. simulating FRAP experiments) ¹¹.
- **Usability:** Modern VCell versions have search/autocomplete for large models, integration with COPASI for parameter estimation, and interactive plotting of results. The GUI was overhauled in VCell 4.0 (2005) to an intuitive multi-document interface ²⁵. Users may run simulations locally or on VCell's compute cluster; in VCell 7.x, containerized solvers (Docker/Singularity) are supported and a command-line interface (CLI) is available.
- **Extensibility:** The codebase is modular (vcell-core, vcell-client, vcell-solvers, etc.) and on GitHub ¹³. Community projects include the **VCell-Fiji** plugin (to link VCell with ImageJ workflows) ¹⁵ and **PyVCell** (Python API for scripting) ¹⁶. A Google Summer-of-Code project **VCell-AI** (2025) is building an AI-driven model explorer.

Related Tools, Forks, and Integrations

VCell does not exist in isolation: it leverages and interoperates with many computational biology tools:

- **Modeling Standards:** SBML (import >90% of curated models ³⁵), SED-ML/COMBINE export (VCell 7.x) for reproducible experiments. VCML (Virtual Cell Markup Language) is a public format, enabling external tool use ²⁷.
- **Solver Engines:** Interfaces to **BioNetGen/NFSim** (rule-based), **Smoldyn** (particle simulation) ⁹, **COPASI** (parameter estimation and rate equation integration) ⁴⁵. VCell acknowledges contributions from third-party solvers (SUNDIALS CVODE, FronTier, etc. ⁴⁶).
- **Pathway Databases:** Integration with **BioPAX/Pathway Commons** allows importing reaction/pathway data into a VCell model ¹¹ ³⁰. VCell 5.x introduced a BioPAX interface (Blinov et al. 2014) ³⁰.
- **Visualization & Analysis:** VCell results can be exported for VisIt or ImageJ; the ImageJ/VCell plugin can fetch simulation data and overlay on images ¹⁵. Tutorials and the **Virtual Cell Playground** (web tool) help newcomers.
- **Forks/Extensions:** Projects like **CompCellBio** (a portal including VCell models) and **SpringSaLaD** (particle simulator for large molecules) were developed by CCAM collaborators. On GitHub, *virtualcell/vcell* is the main repo; others include *vcell-fvsolver*, *vcell-solvers* (C code), *pyvcell*, *libvcell*, *vcell-fiji*, etc. These support niche uses (e.g. vcell-ode repo contains ODE solvers) ⁴⁷.
- **Community Tools:** Over 1900 registered VCell users have contributed >300 published models ¹⁷. The

VCell database itself contains curated models which serve as a valuable *dataset* for benchmarking other tools. For example, VCell was used to reproduce results of published models (e.g. signaling networks, FRAP analysis), and model exchange with COPASI and other platforms is possible. VCell's "test suite" of public models is used for regression testing SBML support ⁴⁸.

Publications and Case Studies

VCell has been documented in numerous peer-reviewed publications, including seminal papers and application studies. Key references include:

Year	Citation (Authors, Title)	Journal/Source
1997	Schaff <i>et al.</i> , "A general computational framework for modeling cellular structure and function."	<i>Biophysical Journal</i> ¹⁹
2001	Loew & Schaff, "Physiological modeling with the Virtual Cell framework."	<i>Trends in Biotechnology</i> (Review) ²³
2008	Moraru <i>et al.</i> , "The Virtual Cell modeling and simulation software environment."	<i>IET Systems Biology</i> ¹
2012	Resasco <i>et al.</i> , "Virtual Cell: computational tools for modeling in cell biology."	<i>Wiley Interdiscip. Rev. Syst Biol Med</i> ²⁸
2012	Cowan <i>et al.</i> , "Spatial modeling of cell signaling networks."	<i>Methods Cell Biol.</i> ⁴⁹
2014	Blinov <i>et al.</i> , "Pathway Commons at Virtual Cell: use of pathway data for modeling."	<i>Bioinformatics</i> ³⁰
2016	Schaff <i>et al.</i> , "Spatial stochastic-deterministic hybrid modeling."	<i>PLoS Comput Biol</i> ⁸
2017	Blinov <i>et al.</i> , "Compartmental and spatial rule-based modeling with VCell."	<i>Biophysical Journal</i> ³²
2021	Korobkin <i>et al.</i> , "Calcium oscillations in human spermatozoa... VCell models."	<i>Int. J. Mol. Sci.</i> (application) ⁵⁰
2025	Nakata <i>et al.</i> , "Microtubule-binding of Tau in living neurons."	<i>Neurosci Res</i> (VCell study) ³⁹
2025	Kanda <i>et al.</i> , "Modeling EGFR-ERK signaling in cancer cells."	<i>Biophys J.</i> (VCell co-authored) ³⁸

These publications illustrate VCell's development and usage. For example, **Moraru et al. (2008)** presents the software architecture and adoption statistics ¹⁷; **Resasco et al. (2012)** reviews VCell's features; and **Blinov et al. (2017)** demonstrates the rule-based interface. Numerous case studies use VCell to analyze specific biological systems – e.g., calcium dynamics in sperm (Korobkin 2021), FRAP in bacterial spores, signaling pathways in stem cells and neurons (2025 *Biophysical Journal*) ³⁹ ⁵¹. VCell models are routinely associated with publications (the VCell website lists public models by DOI ⁵² ⁵³), supporting reproducible computational biology.

Notable Applications

VCell has been applied to a wide range of cell-biological problems, often in collaboration with experimentalists. Examples include:

- **Calcium Signaling:** Early VCell models simulated IP₃-mediated Ca²⁺ waves in neurons and sperm cells, predicting spatial gradients ⁵⁴ ⁵⁰ .
- **Neuroscience:** Modeling of dendritic spine signaling (Oster & Rangamani, 2016 PNAS) and IPSP generation (Jones et al., 2009) used VCell's geometry and kinetics capabilities.
- **Chemotaxis:** Step-by-step VCell tutorials have demonstrated chemotactic network modeling (Bhattacharya & Iglesias, 2016).
- **Cell Polarity & Motility:** VCell was used to link actin dynamics to cell turning (Roy et al., J Cell Biol. 2001) ⁵⁵ .
- **Morphogenesis:** Models of budding yeast or embryo development have leveraged VCell's multicompartment and spatial features.
- **Disease Modeling:** VCell has been used in disease contexts (e.g. Parkinson's tau dynamics ³⁹ , cancer signaling ³⁸).
- **FRAP/Imaging Studies:** The Virtual FRAP module (from COPASI tech in VCell 5.1) allows fitting diffusion coefficients from photobleaching experiments. Recent work (2025) modeled FRAP in *Bacillus* spores ⁵⁶ .
- **Systems-Level Pathways:** The RanGTP gradient (Smith et al. 2002, Science) and platelet coagulation networks (Podoplelova 2016) were built in VCell.

These case studies demonstrate VCell's core strength: integrating quantitative experimental data with biophysical modeling. Over 300 **public models** are available (covering metabolism, signaling, electrophysiology, etc.), making VCell not only a simulator but also a community repository of computational models.

Executive Summary of Key Points

- **Origins & Team:** VCell was conceived by Leslie Loew's CCAM group (UConn) as an NIH-funded modeling resource (est. 1994) ¹⁸ ³ . Core developers include Schaff, Slepchenko, Moraru, Blinov, and collaborators.
- **Functionality:** VCell's GUI simplifies building compartmental/spatial models; it auto-generates mathematical equations and uses high-performance solvers ¹ ⁹ . It supports deterministic, stochastic, hybrid and rule-based simulations.
- **Software Milestones:** Major releases: VCell 4.0 (2005, new UI) ²⁵ ; 5.0 (2011, Smoldyn/segmentation) ⁷ ; 6.0–6.1 (2016–17, rule-based) ⁸ ; 7.0 (2018, modular redesign) ³³ ; 7.x (2019–2023, standards and container support) ³⁴ ³⁶ ; 7.7 (2026, latest release with easy install) ³⁷ .
- **Publications:** Foundational papers (e.g. Schaff 1997, Moraru 2008) set out VCell's framework ¹⁹ ¹ . Recent work covers rule-based modeling (Blinov 2017) and hybrid solvers (Schaff 2016). Reviews summarize capabilities (Resasco 2012) and highlight VCell in systems biology ²⁸ .
- **Integrations:** VCell works with SBML/SEDML standards, BioNetGen (for rules), Smoldyn, COPASI, BioPAX, and more. It features a BioPAX interface (Pathway Commons) ¹¹ ³⁰ and an ImageJ plugin for image-based modeling. The GitHub-hosted code (vcell-core, vcell-server, etc.) invites external development ¹³ ¹⁶ .
- **Community & Resources:** Over 300 public models (with publications) are accessible through the VCell website. Training courses and a mailing list support users. VCell acknowledges its collaborative

ecosystem (COPASI, SBML, etc.) ⁵⁷ . New projects (e.g. VCell-AI, CompCellBio portal) continue to evolve the platform.

Sources: Official VCell website (vcell.org) and documentation ² ⁵⁸ , key papers and reviews ⁵³ ¹ ²⁸ .

Slide Presentation Outline

1. Slide 1: Virtual Cell Overview

2. VCell is an NIH-funded, open-source platform for cell modeling (biochemical + electrophysiological) ¹ ² .

3. Accessible as a web app/Java client; no coding needed for model construction.

4. Supports ODE, PDE, stochastic, particle and hybrid simulations in realistic geometries ⁵ ⁹ .

Speaker notes: Introduce VCell's purpose and scope. Emphasize user-friendly design for biologists.

Visual suggestion: Schematic of VCell workflow (biological model → math → simulation), or screenshot of VCell GUI with model diagram.

5. Slide 2: History & Funding

6. VCell developed at UConn's CCAM (est. 1994) by Loew et al. ¹⁸ . First paper in 1997 Biophys J ¹⁹ .

7. NIH funding history: NCCR (RR13186) initially; currently NIGMS R24 GM137787 ² ³ .

8. Core team: Leslie Loew (PI), James Schaff, Boris Slepchenko, Ion Moraru, Michael Blinov, etc. ⁴⁰ .

Speaker notes: Cover timeline of founding and funding. Highlight continuous NIH support.

Visual suggestion: Timeline chart (genesis of CCAM → key grants) or photo of CCAM team.

9. Slide 3: Timeline of VCell Milestones

10. Bullet-list or table: key dates (1997 initial paper; 2005 VCell 4.0; 2011 VCell 5.0; 2016–18 hybrid & rule-based features; 2018 VCell 7.0; 2023–26 latest releases) ²⁵ ³³ .

11. Note major publications (2008 IET Systems Biol, 2012 WIREs, 2016 PLOS, 2017 Biophys J) aligned to timeline.

Speaker notes: Walk through development chronology and how features evolved.

Visual suggestion: **Mermaid timeline** (as above) showing milestones, or a Gantt chart of releases/papers.

12. Slide 4: Architecture & Workflow

13. VCell has **BioModel (GUI for biology)** and **MathModel (equations)** interfaces ⁴¹ ⁴² . The "Physiology" tree specifies compartments, species, reactions. Applications define simulation conditions (well-mixed, spatial, etc.).

14. Automatic math generation: VCML code and solver assignment.

15. Output: plots, data export, or live visualization.

Speaker notes: Explain the layered abstraction (model → math → simulation) and how VCell encapsulates complexities.

Visual suggestion: Flowchart of VCell pipeline: Physiology → BioModel → MathModel → Solver → Results (mermaid flowchart style).

16. Slide 5: Key Features & Modules

17. **Spatial modeling:** 3D geometries (meshes from images), diffusion, flow ⁵ ¹¹ . 2D/3D membrane diffusion.

18. **Rule-based modeling:** Integrated interface (VCell 6.x); can mix reactions & rules ⁹ .

19. **Analysis tools:** Virtual FRAP fitting (built-in), parameter estimation (via COPASI), global sensitivity analysis.

20. **Standards support:** SBML import/export, BioPAX queries, SED-ML (COMBINE) export ³⁴ ¹¹ .

Speaker notes: Summarize capabilities beyond basic reactions: imaging data, stochastic simulation, rule-based.

Visual suggestion: Collage of screenshots: one of geometric mesh, one of rule-based editor, etc. (Alternatively, icons representing each feature.)

21. Slide 6: Software Releases

22. **VCell 4.0 (2005):** Redesign UI, MDI, database manager ²⁵ .
 VCell 5.0 (2011): New GUI, Smoldyn, BioPAX support ⁷ .
 VCell 6.x (2016–17): Hybrid solver, rule-based GUI.
 VCell 7.0 (2018): Modular core, cloud/HPC support ³³ . Subsequent 7.x added SBML/SEDML support, containerized solvers (7.5 in 2023) ³⁶ .

Speaker notes: Emphasize major software generational changes (UI overhaul, new modeling paradigms).

Visual suggestion: Timeline graphic or table summarizing version vs year vs major features.

23. Slide 7: Integrations & Tools

24. **Community standards:** Uses SBML, SED-ML/OMEX for model exchange. VCell Markup (VCML) is public ²⁷ .

25. **Interoperability:** Interfaces with COPASI (PE), BioNetGen/NFSim, Smoldyn, Sundials, etc. ⁹ ⁵⁷ .

26. **Data sources:** Annotated models (MIRIAM), links to PubMed/UniProt; Pathway Commons/BioPAX querying (2014) ³⁰ .

27. **Extensions:** PyVCell (Python API), VCell-Fiji plugin ¹⁵ ¹⁶ . GitHub repos for vcell-solvers, vcell-fvsolver, etc.

Speaker notes: Show VCell's ecosystem compatibility. Highlight open-source code and plugins.

Visual suggestion: Block diagram of VCell in center with arrows to standards (SBML, COMBINE), tools (COPASI, BioNetGen), and databases (UniProt, Pathway Commons).

28. Slide 8: Publications and Validation

29. **Key papers:** Schaff *et al.* 1997 (*Biophys J*) – first VCell framework ¹⁹ ; Moraru *et al.* 2008 (*IET Syst Biol*) – architecture and adoption ¹⁷ ; Resasco *et al.* 2012 – VCell review ²⁸ ; Schaff *et al.* 2016 (*PLoS CB*) – hybrid solver; Blinov *et al.* 2017 (*Biophys J*) – rule-based.

30. **Use cases:** Citing VCell is encouraged (Publish a model guidelines) ⁵³ . ~300 published models with DOIs.
31. **Benchmarks:** VCell's extensive model library provides test cases for simulation tools.
Speaker notes: Highlight the influential publications and community adoption. Mention model database as a resource.
Visual suggestion: Table listing 3–4 landmark papers (as shown above), or a chart of number of publications/models over time.
32. **Slide 9: Case Studies & Applications**
33. Examples: **Calcium signaling** models (Korobkin 2021) ⁵⁰ ; **Chemotaxis** networks (Bhattacharya 2016); **Spore FRAP** (Kanda 2025) ⁵⁶ ; **Stem cell regulation** (Drosophila testis, 2025) ⁵⁶ .
34. Multidisciplinary use: Experimentalists use VCell for hypothesis testing (e.g. predicting diffusion coefficients from imaging).
Speaker notes: Give 1–2 compelling examples of how VCell models yielded biological insights.
Visual suggestion: Diagram from a paper (e.g. neuron illustrating calcium wave) or a snapshot of a published VCell model (with permission), plus brief caption.
35. **Slide 10: Community & Future Directions**
36. **Open Source & Training:** VCell's GitHub hosts code; active mailing list and workshops (e.g. CCAM courses).
37. **Collaborations:** Annual presentations at cell biology conferences; integration with the Center for Reproducible Biomedical Modeling ⁵⁹ .
38. **Emerging work:** VCell-AI (model discovery), expanding multicellular modeling, GPU acceleration (unpublished/upcoming).
39. **Summary:** VCell continues evolving (e.g. focus on reproducibility, standardization, ease of use). The ecosystem (SBML, COPASI, BioModels) ensures interoperability.
Speaker notes: Close by emphasizing VCell's community-driven growth and alignment with FAIR modeling practices.
Visual suggestion: Logos of VCell partners (COPASI, SBML, etc.) or a slide with "Looking Ahead" bullet points.

¹ ⁵ ⁶ ¹² ¹⁷ ²⁷ ⁴⁰ (PDF) Virtual Cell modelling and simulation software environment

https://www.researchgate.net/publication/224338401_Virtual_Cell_modelling_and_simulation_software_environment

² ⁴ ¹⁰ ¹⁸ ³⁷ ³⁸ ³⁹ ⁴⁶ ⁵⁰ ⁵¹ ⁵² ⁵⁶ ⁵⁷ ⁵⁹ VCell- Modeling & Analysis Software – Virtual Cell Modeling & Analysis Software

<https://vcell.org/>

³ ²² ²³ ⁵⁴ The Virtual Cell: a software environment for computational cell biology - ScienceDirect

<https://www.sciencedirect.com/science/article/abs/pii/S0167779901017401>

⁷ ⁹ ¹¹ ¹⁵ ²⁵ ²⁶ ³³ ³⁴ ³⁵ ³⁶ ⁴⁸ ⁵⁸ Download VCell Software – VCell- Modeling & Analysis Software

<https://vcell.org/run-vcell-software>

8 31 44 VCell News & Events – VCell- Modeling & Analysis Software

<https://vcell.org/vcell-news-events>

13 GitHub - virtualcell/vcell: Virtual Cell Framework · GitHub

<https://github.com/virtualcell/vcell>

14 16 47 Virtual Cell · GitHub

<https://github.com/virtualcell>

19 29 32 49 53 Publish A VCell Model – VCell- Modeling & Analysis Software

<https://vcell.org/publish-a-vcell-model>

20 21 24 30 55 VCell Publications – VCell- Modeling & Analysis Software

<http://vcell.org/vcell-publications>

28 Virtual Cell: computational tools for modeling in cell biology - PubMed

<https://pubmed.ncbi.nlm.nih.gov/22139996/>

41 42 43 45 About VCell – VCell- Modeling & Analysis Software

<https://vcell.org/why-vcell>